

TRUSTROBOTICS

BRAZILIAN JIU-JITSU ROBOT

Interactive Physical Intelligence for Humanoid Robots

A reference architecture for multimodal relational-state reasoning,
closed-loop grappling action, submission protocols, persona policies, and
independent safety enforcement.

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Related Patent: DOR-282

Brazilian Jiu-Jitsu Humanoid Robot Physical Action Control

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Executive Summary

Brazilian jiu-jitsu is an unusually demanding test of physical intelligence. Two independently acting bodies remain in continuous contact while their grips, frames, hooks, posts, load paths, balance, leverage, and available actions change from moment to moment. Much of the decisive physical information is occluded from cameras, and two interactions that share the same position name may present materially different mechanical realities.

The TrustRobotics BJJ Robot architecture addresses this problem through a closed-loop system that combines multimodal sensing, a BJJ interaction model, an optional inter-body relational-state representation, physical-action generation, and actuator control. The robot does not simply recognize a position and replay a memorized motion. It senses the interaction, determines the current physical state, acts, observes the opponent response, updates its state, and selects the next action.

CENTRAL THESIS A humanoid should reason about changing relationships between bodies—not merely recognize bodies as objects or classify named positions.

The architecture also treats submission behavior and safety as first-class protocol elements. A robot should recognize human taps and verbal submissions, promptly release or reduce force, and concede when a human has correctly established a submission—even if the robot’s mechanical construction would tolerate the position. Competitive behavior, aggressiveness, and persona are subordinate to an independently enforced safety layer.

1. The Physical-Intelligence Challenge

Conventional robot perception often assumes that objects can be separately observed, classified, and manipulated. Grappling breaks that assumption. The robot and its counterpart form a coupled physical system in which contact itself carries information and constrains future action.

- A wrist grip, forearm frame, leg hook, hand post, underhook, or pin can determine what action is physically available.
- The same semantic position can contain different load paths, joint alignments, pressure distributions, and escape routes.

- Relevant body regions may be hidden, requiring tactile, force, torque, pressure, proprioceptive, inertial, and audio sensing in addition to vision.
- An opponent's defense may simultaneously defeat one technique and create the entry condition for another.

For these reasons, image recognition alone is insufficient. The central engineering problem is maintaining a machine-usable interaction state that reflects the changing relationships between both bodies and supports responsive physical action.

2. Reference Architecture

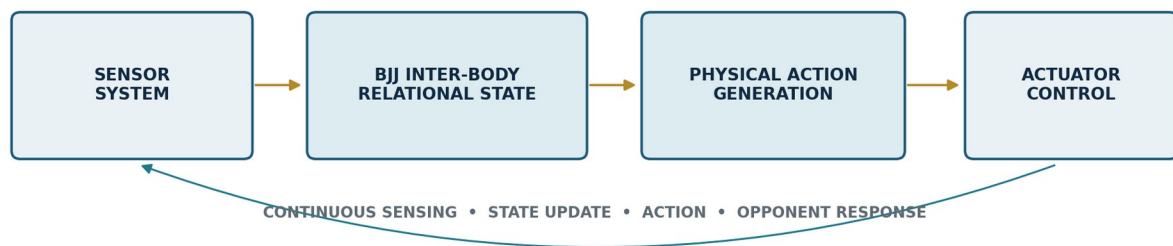


Figure 1. Reference runtime architecture and continuous interaction loop.

The reference architecture contains four principal runtime components. Implementations may combine components, distribute them across computing systems, or use alternative learned or symbolic structures, provided that the closed-loop behavior is maintained.

2.1 Sensor System

Collects available information concerning the robot, opponent, and environment. Inputs may include RGB and depth cameras, tactile arrays, force and torque sensors, pressure sensing, joint state, proprioception, inertial measurements, proximity sensing, and microphones.

2.2 BJJ Interaction Model

Processes sensor information to determine a machine-usable state of the interaction. The state may include position, technique, technique progression, participant response, interaction history, safety state, physical conditions, and inter-body relationships.

2.3 Physical Action Generation

Receives BJJ-specific machine information and generates a technique, technique component, action primitive, trajectory, target pose, force or torque objective, control objective, physical-action token, or actuator-level information. A vision-language-action model is one possible pathway, but it is not required.

2.4 Actuator Control

Executes the selected physical action within mandatory constraints. Execution remains observable so the system can detect opponent response, update state, and alter or terminate the action.

3. Inter-Body Relational State

A preferred representation describes physical relationships between body regions of the robot and opponent. This representation allows the system to reason below the semantic level of conventional BJJ position names.

Relationship class	Examples	Machine significance
Contact and control	contact, grip, hook, pin, entanglement	Identifies constrained regions and available control surfaces.
Structure and support	frame, post, support, base	Represents resistance, balance, and paths that prevent movement.
Mechanics	load, force transmission, pressure, constraint	Captures how force and weight pass through the coupled bodies.
Geometry and motion	relative orientation, relative movement	Supports transition detection and prediction of reachable states.

A relational state may be implemented as a graph, hypergraph, matrix, tensor, structured state, embedding, token sequence, latent representation, or another machine-usable structure. For example, ROBOT_LEFT_HAND — GRIP → OPPONENT_RIGHT_WRIST may be more actionable than classifying the exchange merely as “guard.”

4. Positions, Techniques, and Transformations

The initial positional ontology should recognize standing, guard, side control, mount, and back mount, with extension to closed guard, open guard, half guard, turtle, knee-on-belly, north-south, front headlock, clinch, and leg entanglement. Named positions remain useful abstractions, but the underlying relational state should remain available.

Techniques are preferably represented as transformations rather than fixed human trajectories:

ENTRY RELATIONAL STATE → INTERMEDIATE STATE(S) → TARGET RELATIONAL STATE

A sweep, for example, may be defined by controlling selected body regions, removing a post or base, redirecting weight, rotating or translating the opponent, and establishing top control. The robot joint trajectory may differ from the motion demonstrated by a human while achieving the same physical relationships and objective.

5. Closed-Loop Action and Decision Policy

BJJ is modeled as a continuing exchange rather than a sequence of isolated commands. Each cycle begins from the physical state produced by the previous action and the opponent's response.

INTERACTION EQUATION $\text{Action}_n \rightarrow \text{Opponent Response}_n \rightarrow \text{State}_{n+1} \rightarrow \text{Action}_{n+1}$

The robot may maintain a technique decision architecture associating current state, objective, available techniques, anticipated response, counter or transition, and resulting state. This architecture may be implemented as a decision graph, learned policy, transformer, reinforcement-learning policy, planner, neural model, rules structure, or hybrid. Participant-response prediction and world or physics models may improve selection but are optional.

6. World and Physics Models

When present, a world or physics model may represent center of mass, base of support, momentum, force, torque, leverage, friction, pressure, load distribution, joint loading,

range of motion, balance, participant strength, and robot actuator capability. Candidate techniques may then be evaluated according to predicted future physical or relational states. The architecture does not require a physics model merely to generate a BJJ action.

7. Submission Protocol

7.1 Opponent Submission

Submission recognition should preempt ordinary technique execution. Signals may include tapping the robot, tapping the mat, repeated tapping, verbal “tap,” a designated gesture, or another configured indication. Once detected, the robot should promptly release or reduce the relevant physical action.

7.2 Robot Submission

A humanoid should be capable of tapping out when a human has correctly established a submission, even if the position cannot mechanically damage the robot. Concession may be communicated by physically tapping, tapping the mat, verbal output, an electronic signal, cessation of resistance, or another agreed signal. This preserves the semantic integrity of training and prevents the robot from exploiting non-human joint limits.

8. Independent Behavior Controls

Control	Primary effect	Constraint
Aggressiveness	Attack frequency, pace, pressure, persistence, transitions, positional risk, control-versus-attack preference.	Cannot exceed mandatory safety limits.
Tap sensitivity	Threshold at which the robot recognizes a successful submission against itself and concedes.	Independent of aggressiveness.
Safety envelope	Force, torque, pressure, joint, biomechanical, distress, and other execution limits.	Supersedes all behavioral settings and personas.

An attacking robot can therefore remain conservative about submissions—for example, aggressiveness 8/10 and tap sensitivity 9/10—while all behavior remains bounded by the independent safety envelope.

9. Persona Policy Cards

The same humanoid hardware may express different BJJ decision styles through downloadable or selectable persona policy cards. A card may represent characteristics associated with a practitioner, instructor, academy, lineage, methodology, skill level, or synthetic persona.

- Positional priorities, preferred techniques, technique chains, and response mappings.
- Timing, pressure, aggressiveness, tap sensitivity, and control-versus-attack preferences.
- Decision rules, embeddings, adapters, model parameters, and technique weights.
- Demonstrations, corrections, relational-state sequences, decisions, and outcomes learned through direct practitioner training.

The objective is not necessarily to reproduce a practitioner’s exact motion. The system may instead learn the practitioner’s decision relationships: which objectives and transitions are preferred from particular physical states. No persona may override a mandatory safety constraint.

10. Training and Deployment Modes

Mode	Purpose
Instruction	The robot demonstrates or teaches a technique.
Drilling	The robot repeatedly provides a configured reaction or position.
Adaptive training	Resistance and reactions change according to student performance.
Rolling	Continuous interactive BJJ with closed-loop state updates.
Competition simulation	A selected ruleset, skill level, persona, and strategy govern the exchange.
Robot-versus-robot	Humanoids train through self-play or competitive interaction.

11. Safety Architecture

Competitive behavior and physical safety must remain separable. The safety layer should be capable of limiting, modifying, or terminating an action based on force, torque, pressure, joint state, biomechanical condition, participant distress, sensor confidence, or another safety condition regardless of the active persona or competitive objective.

NON-NEGOTIABLE RULE Safety constraints supersede persona, aggressiveness, tap sensitivity, and learned action policy.

A practical validation program should combine component tests, simulated contact, instrumented dummies, controlled human-in-the-loop trials, redundant tap channels, emergency-stop behavior, conservative force escalation, event logging, and post-interaction review. The milestone is not simply executing techniques; it is executing, updating, conceding, and releasing predictably.

12. Applications Beyond BJJ

The same physical-interaction architecture may support no-gi grappling, wrestling, judo, sambo, mixed martial arts, rehabilitation, physical instruction, assisted mobility, collaborative manipulation, and other contact-rich interactions. BJJ is valuable as a development environment because it exposes the core problem at high density: continuous reasoning about contact, constraint, force, relative pose, and another intelligent participant's response.

13. Research Agenda

- Determine which relational-state encoding best balances observability, generalization, and real-time control.
- Establish how BJJ knowledge should be divided between the interaction model and action-generation model.
- Measure transfer efficiency from human demonstrations to humanoid-specific trajectories.
- Develop robust fusion of tactile, force, proprioceptive, inertial, audio, and visual information under occlusion.

- Train and validate persona policies without allowing style parameters to weaken safety guarantees.
- Define submission-state confidence, preemption latency, and safe-release acceptance criteria.
- Study how predictive physical models and mandatory constraints interact with learned policies.

14. Initial Demonstration Roadmap

	Capability	Acceptance demonstration
1	Recognize	Identify five fundamental BJJ positions using multimodal perception.
2	Represent	Maintain a machine-usable interaction state, including physical relationships where available.
3	Choose	Recognize several techniques and select a context-appropriate robot action.
4	Respond	Observe a human counter, update the state, and select a subsequent action.
5	Concede	Recognize a human tap and release safely; recognize when the robot has been caught and tap out.

Completing this loop would demonstrate more than robotic martial arts. It would demonstrate a humanoid capable of reasoning and acting through continuously changing physical relationships with another intelligent body.

Conclusion

A useful BJJ robot is not defined by the number of techniques it can replay. It is defined by whether it can understand a coupled physical interaction well enough to choose, execute, revise, and safely terminate actions as the other participant responds. The architecture described here provides a pathway from multimodal contact sensing to relational-state reasoning, closed-loop physical action, configurable training behavior, and independently enforced safety.

BJJ supplies a difficult but productive benchmark for general physical intelligence. Success requires a robot to understand not only where bodies are, but how they are

connected, constrained, loaded, moving, and responding—and to use that understanding continuously.

Patent and Publication Notice

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